

Colonoscopy Polyps Segmentation Using ResUNet and Kvasir-SEG Dataset

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1 Introduction

Colorectal cancer is one of the leading causes of cancer-related deaths worldwide, and early detection of polyps through colonoscopy plays a crucial role in prevention. Manual polyp identification and segmentation during endoscopy can be time-consuming, subjective, and prone to error. Automated segmentation methods using deep learning provide accurate, real-time support to clinicians. This project builds a deep learning model for polyp segmentation using public colonoscopy datasets. We performed exploratory data analysis (EDA), hypothesis testing, preprocessing, and augmentation strategies to train a ResUNet model with a ResNet34 encoder for robust segmentation performance.

2 Dataset

The primary dataset is **Kvasir-SEG** (1,000 images and masks), supported by CVC-ClinicDB (612 images) and the Merged Polyp Segmentation Dataset (MPSD, 22,330 images). These open-access datasets provide diverse polyp appearances and lighting conditions, ensuring robust generalization.

3 Exploratory Data Analysis

We analyzed image dimensions, aspect ratios, resolutions, polyp coverage, and pixel intensities. Findings included:

- Most images are small ($\sim 250 \times 300$ pixels) and nearly square.

- Polyp coverage is skewed toward low values (mean coverage slightly above 0.1).
- Larger polyps dominate the dataset (80%), while small polyps are underrepresented.

4 Hypothesis Testing

- Spearman correlation between resolution and polyp size: 0.169 (weak but significant).
- T-test on portrait vs landscape orientation: $p = 0.028$ (small but significant difference).
- Pearson correlation between polyp size and contrast: 0.081 (very weak correlation).

5 Preprocessing and Augmentation

Images and masks were padded to squares, resized to 256×256 , and normalized using ImageNet statistics. Binary masks were generated for segmentation. Data augmentation included flips, rotations, brightness/contrast adjustments, HSV shifts, Gaussian blur, and translations to improve generalization.

6 Baseline Model

The baseline **ResUNet** architecture consists of:

- Encoder: Convolutional layers with max-pooling.
- Decoder: Upsampling layers with skip connections.
- Loss function: Dice loss ($L_{total} = L_{Dice}$).

Results: Validation Dice = 0.94, IoU = 0.86, Validation Loss = 0.04 after 20 epochs.

7 Enhanced Model

We enhanced the baseline using:

- **ResNet34 encoder** (pretrained on ImageNet) for stronger feature extraction.
- Hybrid loss function: $L_{total} = L_{BCE} + L_{Dice}$ with class weighting.
- Optimizer: AdamW with learning rate scheduling.

Improved Results: Validation Dice = 0.9587, IoU = 0.9541, Validation Loss = 0.0682. The enhanced model produced sharper boundaries, smoother contours, and fewer false positives.

8 Results

Key findings:

- Subtle dataset biases exist (resolution, orientation, contrast) but do not dominate performance.
- Enhanced ResUNet outperformed baseline in Dice and IoU metrics.
- Training curves confirmed stable learning without overfitting.

9 Conclusion

The enhanced ResUNet with a ResNet34 encoder and hybrid BCE + Dice loss significantly improved polyp segmentation performance. Transfer learning and tailored loss functions proved effective for imbalanced medical datasets. The model’s strong accuracy and qualitative outputs demonstrate its potential for real-world clinical applications and future research.